

Introduction

The EHSM (EcoHydrological Streamflow Model) is a conceptual lumped model aimed to daily streamflow simulation. The model, processing daily rainfall and reference evapotranspiration at the basin scale, reproduces surface and subsurface runoff, soil moisture dynamics and actual evapotranspiration fluxes. The key elements of this numerical model are the soil bucket, where rainfall, evapotranspiration and leakage drive soil moisture dynamics, and two linear reservoirs working in parallel with different characteristic response times. The surface reservoir, able to simulate the fast response of the basin, is fed by rain falling on impervious area and by runoff generated with excess of saturation mechanism while the deep reservoir, which simulates the slow response, is fed by instantaneous leakage pulses coming from the soil bucket. The model has seven parameters, which summarize soil, vegetation and hydrological catchment properties. Parameters are assessed performing Monte Carlo simulations.

The model has been here calibrated for three semi-arid river basins located in Sicily, Italy with area ranging from 10 up to 1780 Km² with the aim of investigating how the spatial scale may influence model performances. At the same time, the link between knowledge driven parameters and the calibrated ones is explored, investigating the suitability of a lumped framework for the model as the basin size increases.

1. Model scheme

	Small	Medium	Large
nZr [mm]	620	196	183
s _{fc} [-]	0.895	0.757	0.850
s _w [-]	0.688	0.091	0.158
K _c [-]	1.678	2.264	0.930
K _{sub} [1/day]	0.060	0.346	0.213
K _{sup} [1/day]	0.733	0.865	0.765
c ₀ [-]	0.160	0.088	0.076

Calibrated model parameters.

nZr: active soil depth in the vegetated basin area. Determines the soil water storage capacity.

s_{fc}: soil field capacity. Determines the occurrences of saturation excess.

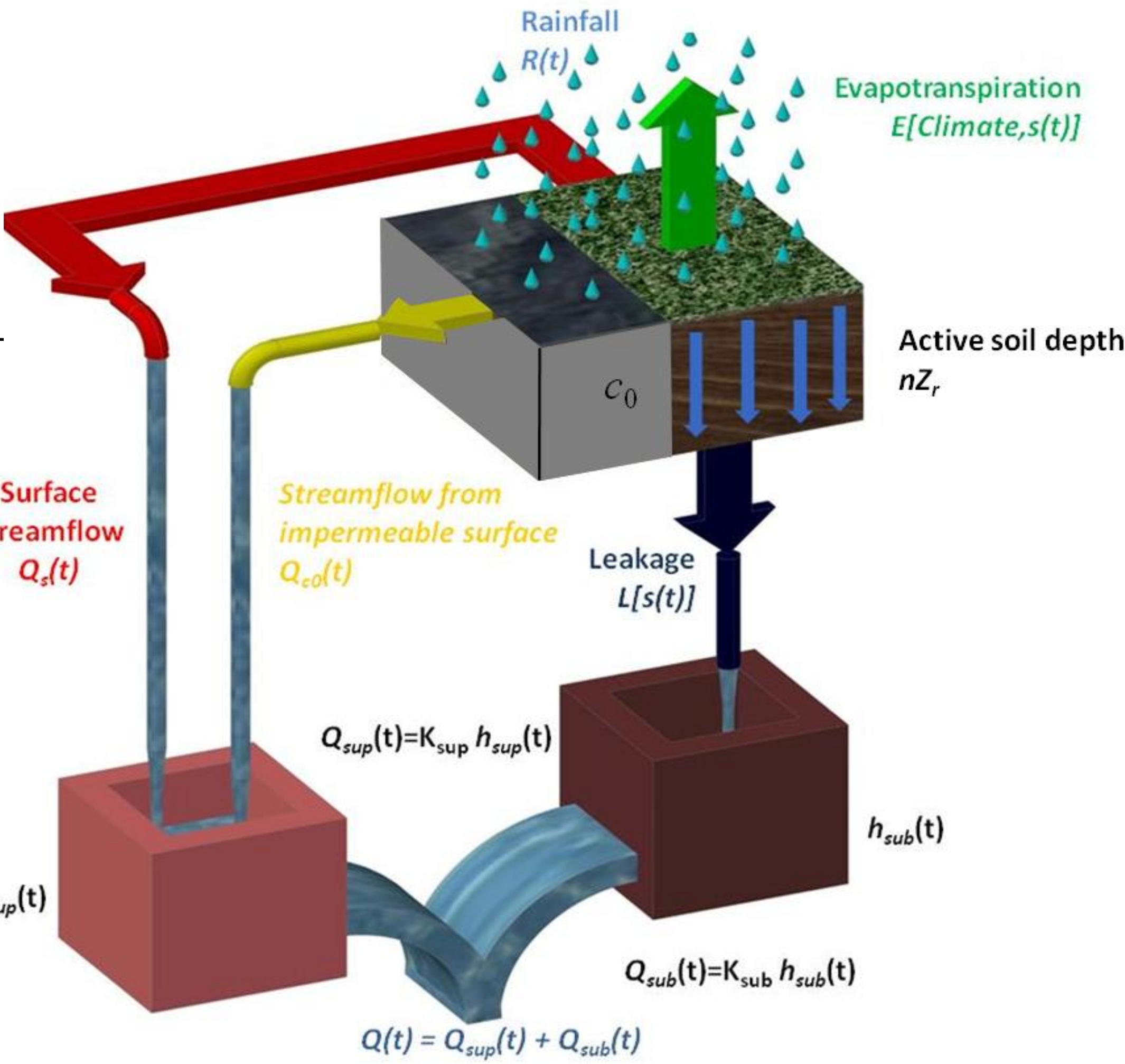
s_w: wilting point. Stops evapotranspiration fluxes.

K_c: Cultural coefficient. It scales the reference evapotranspiration taking into account the real vegetation lying in the basin

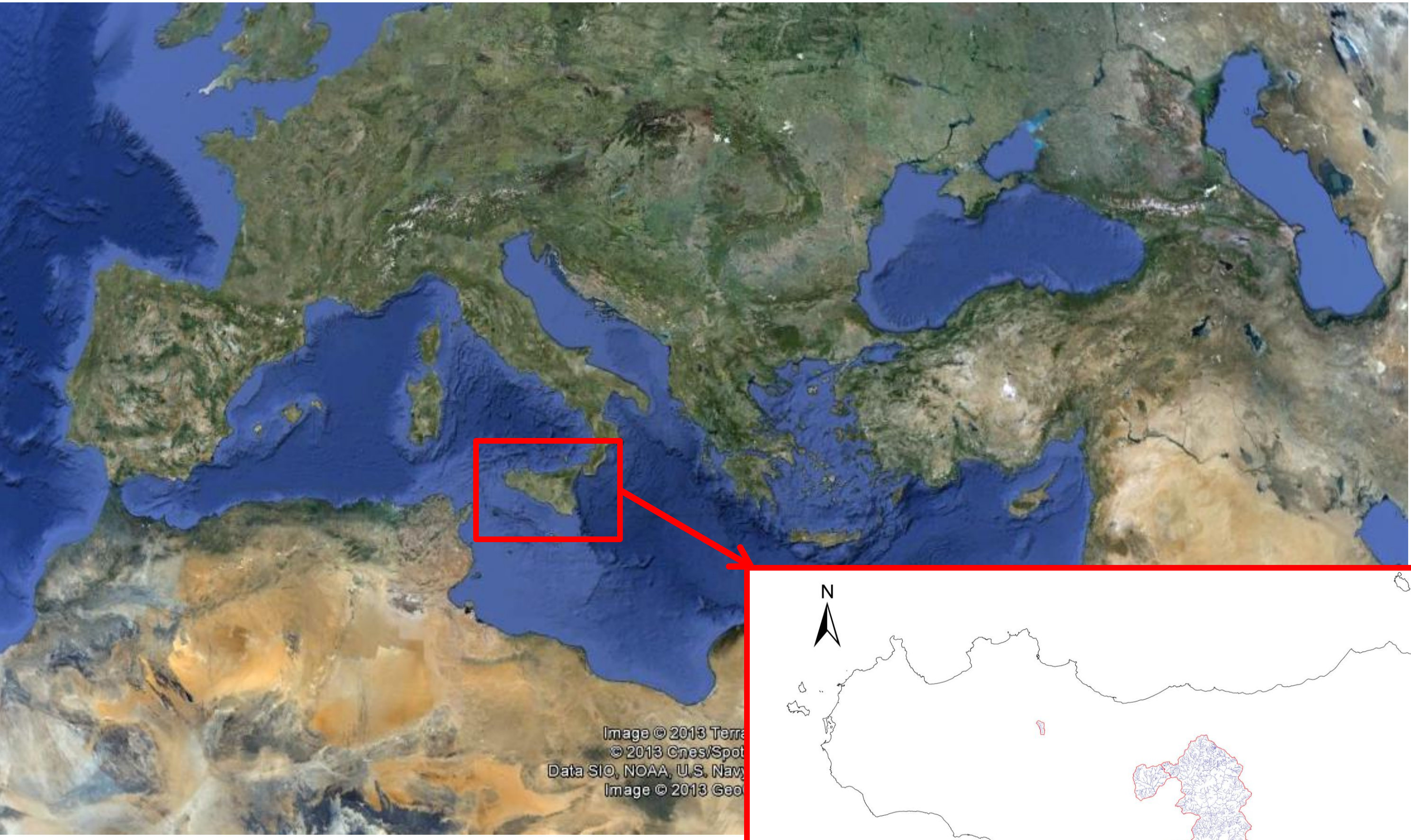
K_{sub}: coefficient representing the inverse of residence time in the sub-superficial reservoir

K_{sup}: coefficient representing the inverse of residence time in the superficial reservoir

c₀: fraction of impervious area



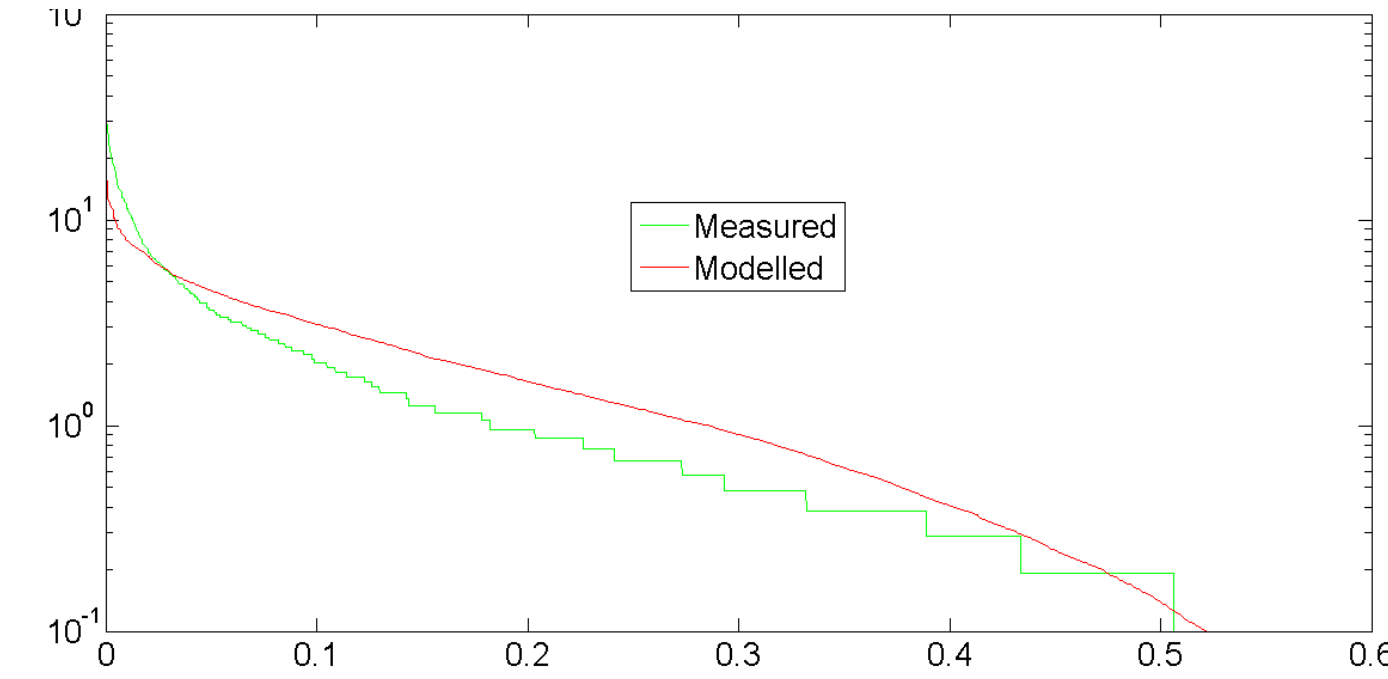
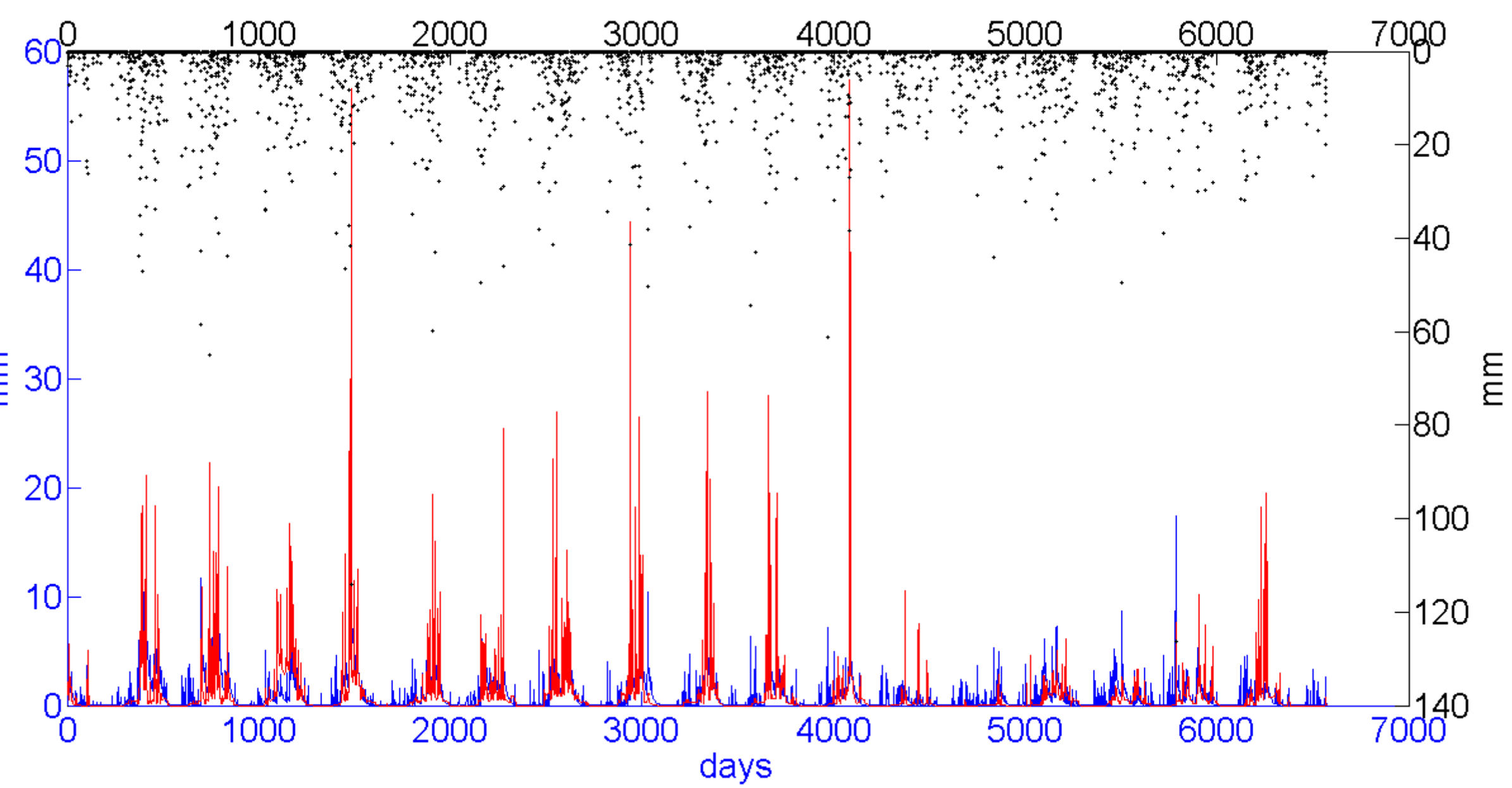
2. Case Study: 3 Sicilian basins



Three basins has been monitored for a long time with regard to temperatures, precipitations and discharges at the basin outlet and, thus, consistently long data series are available.

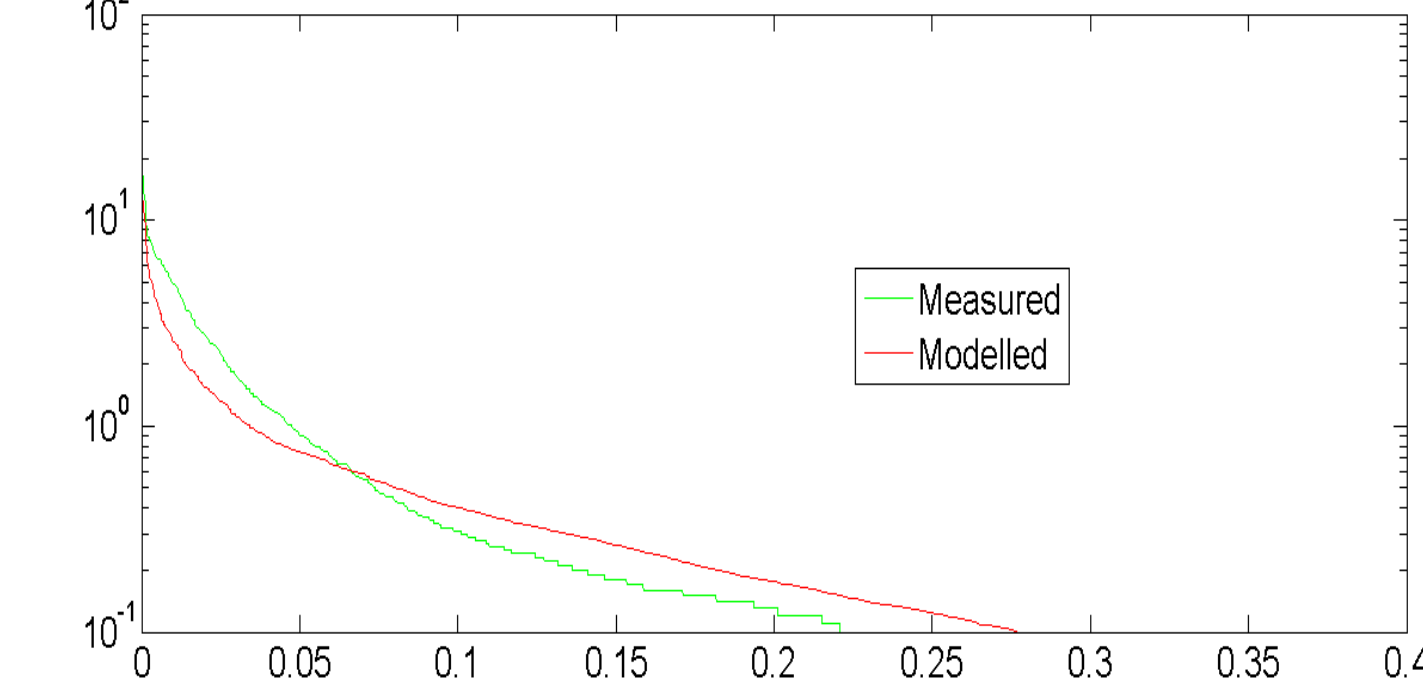
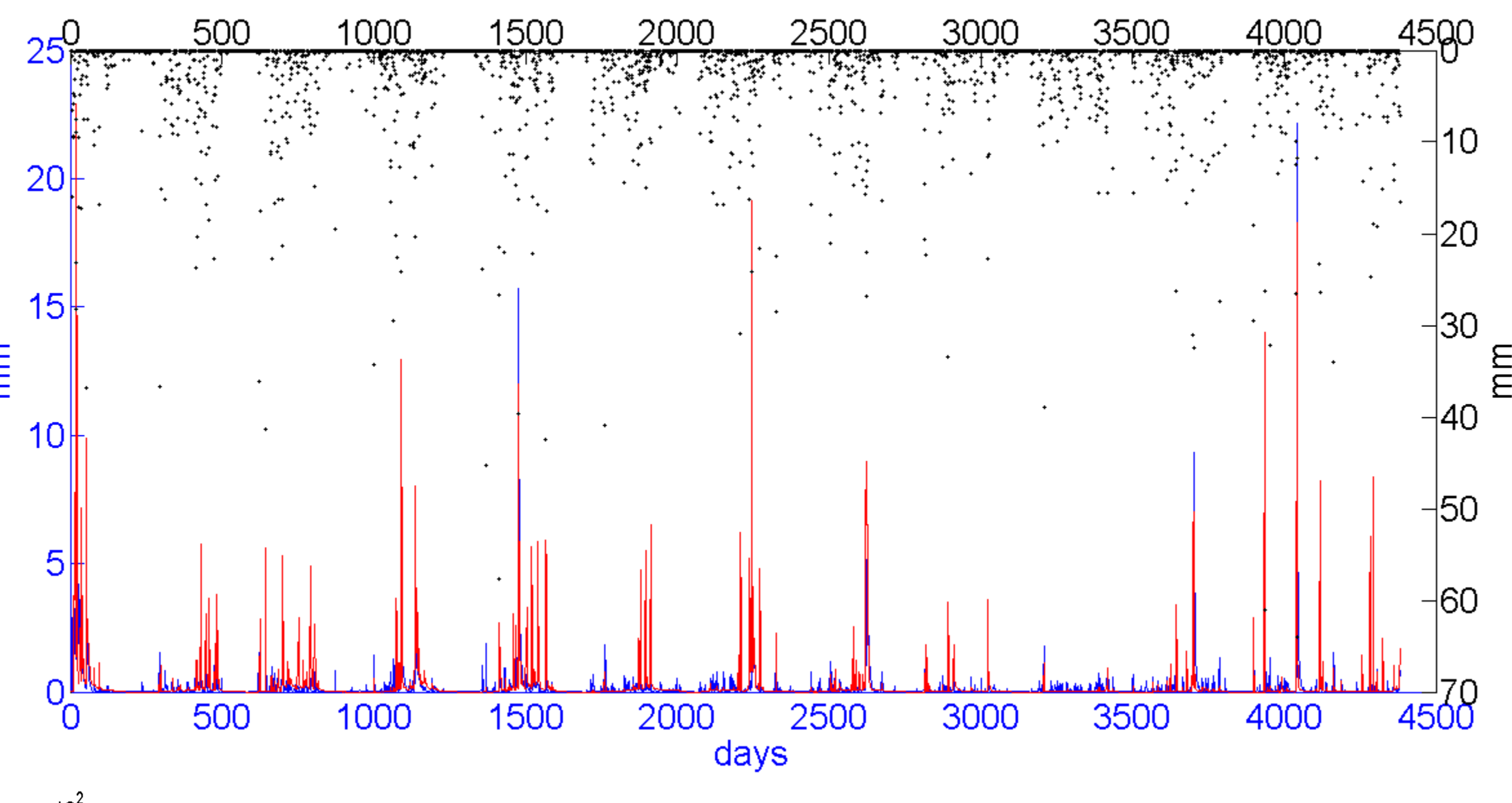
3. Model Application: Results and Discussion

Small basin: 10 Km², Nash-Sutcliffe Efficiency = 0.41



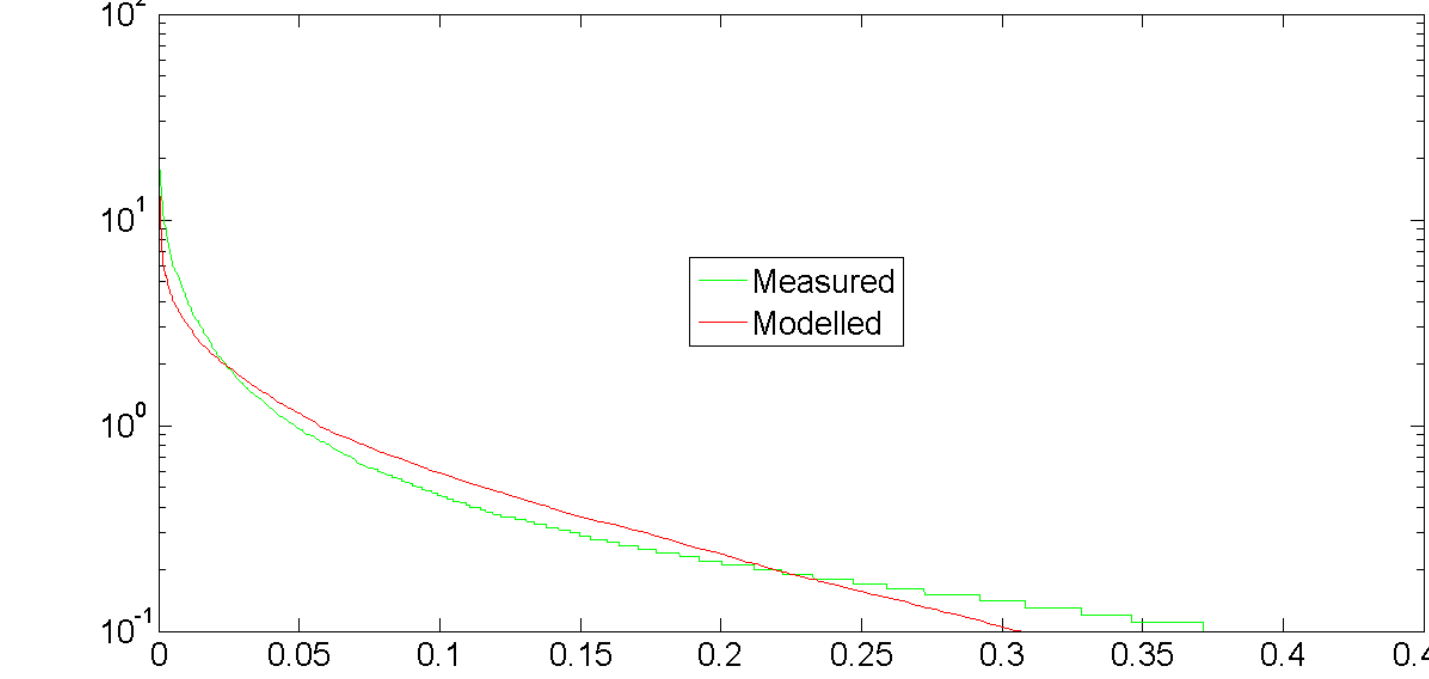
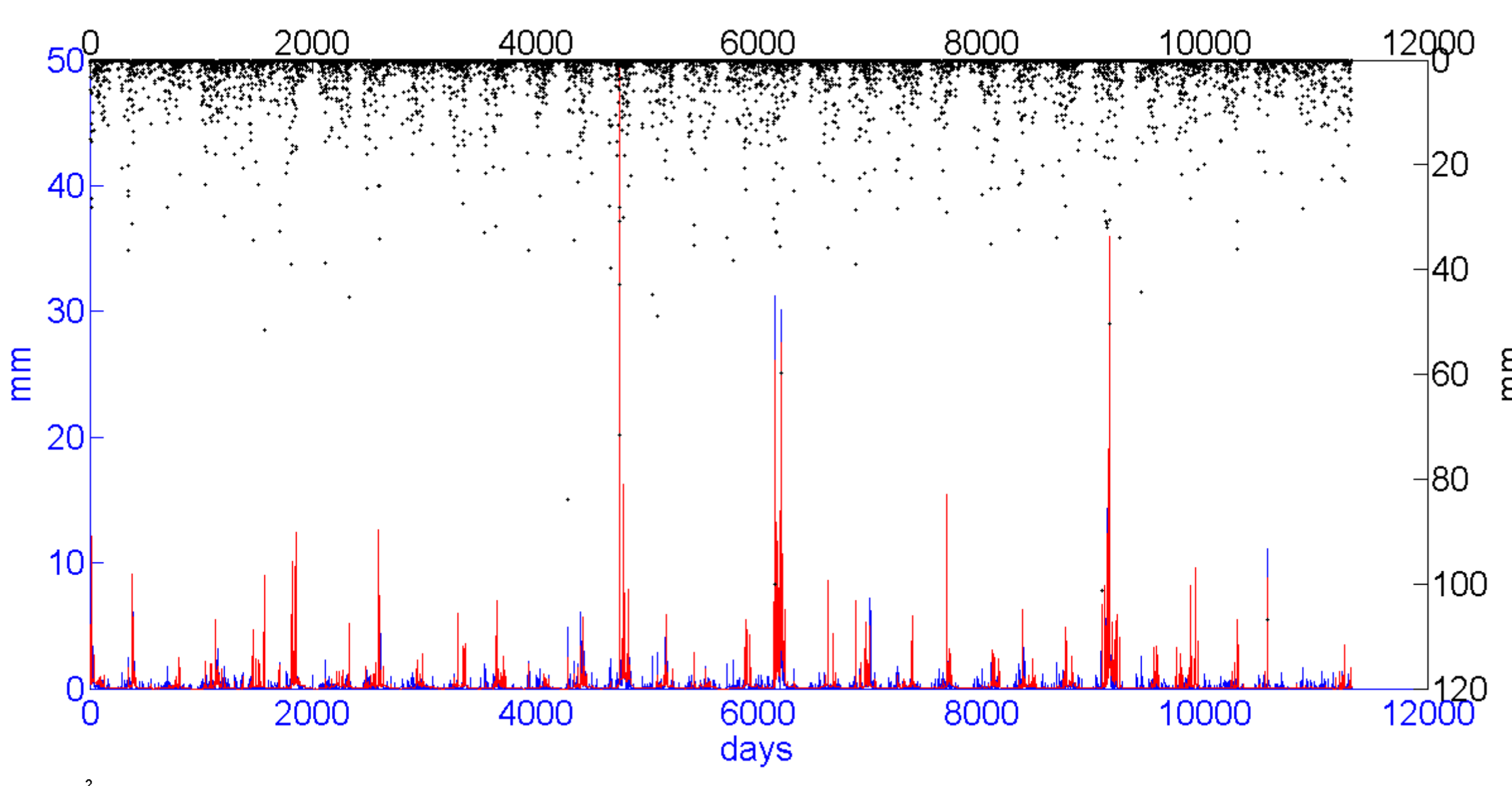
The small basin is characterized by the presence of a forest, which justifies the high value of calibrated active soil depth. The water balance show no saturation excess runoff. Notwithstanding the ephemeral behavior it has the lower duration of dry period among the considered basins. This implies the higher residence time in the sub-surface reservoir. The model performance is the lower of the three considered cases.

Medium basin: 130 Km², Nash-Sutcliffe Efficiency = 0.48



The medium size basin has a marked ephemeral behavior, with a dry period that lasts for the 80% of the year. This has lead to small residence times in both the linear reservoirs. The small soil water storage capacity determines excess of saturation runoff; the high cultural coefficient makes the soil drying process very fast. The model performance is intermediate between the small and the large basin cases.

Large basin: 1780 Km², Nash-Sutcliffe Efficiency = 0.62



The large basin is characterized also by the largest time series. Although one could expect lower performances (because of the higher number of data to reproduce) the model performs better than in the other cases. The model interprets the basin as a shallow active soil, thus producing excess of saturation runoff. The small percentage of impervious area (the lowest among the considered basins) is reflected in a coherent c_0 coefficient.

Conclusions:

The higher the basin area, the higher the model performances. This is probably due to a trade off between data uncertainty and basin size. This is particularly true for streamflow measurement.

Model parameters are driven by the basin hydrological behavior. At the same time they reflect morphological characteristics, and vegetation influence on soil moisture dynamics, which in turns affect runoff generation.

The work has been partially funded by the flagship Project RITMARE - The Italian Research for the Sea - coordinated by the Italian National Research Council and funded by the Italian Ministry of Education, University and Research within the National Research Program 2011-2013.